

Periodic Report for ParkFlow Project

ParkFlow - Smart Parking, Automatic Payment & Urban Mobility Platform

1. Problem Research & Market Study

During the first stage of the project, we focused on understanding the real problems associated with traditional parking systems and identifying the weaknesses of existing solutions.

A. Parking Problem Analysis

- **Traditional Parking Meters:** We studied the limitations of physical parking meters, including installation cost, maintenance requirements, hardware failures, and the inconvenience of requiring users to interact with a separate device.
- **Manual Payment Problems:** Drivers may forget to pay, underestimate parking duration, or need to return to the meter to extend their parking session.
- **Infrastructure Cost:** Installing cameras, sensors, or dedicated hardware for every parking space can significantly increase deployment cost.
- **Scalability:** The solution was designed to reduce the need for expensive hardware so that municipalities can expand the system gradually.

B. Existing Smart Parking Research

We researched different smart parking approaches used internationally, including:

- Parking meters.
- License plate recognition.
- Parking cameras.
- GPS-based parking applications.
- Bluetooth-based vehicle detection.
- Parking sensors.
- Mobile payment platforms.
- The study helped us identify that relying on a single technology could create reliability, cost, or fraud problems.

C. Project Direction

Based on the research, ParkFlow was designed as a mobile-first smart parking platform that combines software-based parking detection, automatic payment, location information, and urban mobility features instead of depending completely on physical parking meters.

2. Smart Parking Architecture & Automatic Payment

The main technical goal is to make the parking process as automatic as possible while minimizing infrastructure requirements.

A. Parking Session Detection

- The mobile application uses the phone's location services to detect when the user reaches a supported parking zone.
- The system analyzes whether the user has stopped inside a registered parking area.
- A parking session can then be initiated through the application.
- The parking duration is recorded automatically until the parking session ends.

B. Automatic Billing

- The user's account is connected to the ParkFlow platform.
- Parking cost is calculated based on the parking zone, parking duration, and municipality pricing rules.
- Instead of manually estimating parking time in advance, the driver pays according to the actual parking duration.
- The architecture supports future integration with digital wallets, credit/debit cards, subscription plans, and municipality payment systems.

C. Offline Connectivity Considerations

- GPS positioning can continue operating without an active internet connection.
- Important parking-session information can be temporarily stored locally on the device.
- Once connectivity returns, the application can synchronize the session with the backend.
- Server timestamps and validation rules can later be used to detect suspicious changes or manipulated sessions.

3. Parking Verification & Anti-Fraud Study

During the architecture discussion, several possible vulnerabilities were identified and evaluated.

A. False Parking Detection

A user could stop near a parking area without actually parking a vehicle. To reduce false detection, the system is designed around multiple signals rather than GPS alone.

- GPS location.
- Movement speed.
- Duration of stationary behavior.
- Entering a known parking geofence.
- Vehicle Bluetooth connection when available.
- Device motion information.
- Parking-session behavior.

B. Bluetooth Limitations

- Some vehicles do not support Bluetooth.
- Users can disable Bluetooth.
- Bluetooth should therefore not be treated as the only verification mechanism.
- ParkFlow uses Bluetooth as an optional supporting signal, not as the core parking system.

C. Camera-Based Verification

- Camera and license-plate recognition systems were studied as a strong verification option.
- Installing cameras at every parking location is expensive.
- Some streets do not already have suitable camera infrastructure.
- Camera installation and maintenance reduce scalability.
- ParkFlow does not require a dedicated camera for every parking space; existing cameras can be used as an optional municipality-side validation layer.

D. Multi-Signal Verification

The proposed design combines several indicators to create a parking-confidence decision instead of depending on a single sensor.

- User inside parking zone.
- Vehicle-speed transition from driving to stationary.

- Stationary duration.
- Device movement pattern.
- Bluetooth vehicle connection if available.
- Parking history and abnormal behavior detection.

4. Smart Parking Zones & Geofencing

A. Parking Zone Mapping

- Zone ID.
- Geographic boundaries.
- Parking price.
- Allowed parking hours.
- Maximum parking duration.
- Municipality information.
- Parking restrictions.

B. Geofencing

The mobile application can detect when the user enters or exits a supported parking zone. This provides the foundation for parking-session suggestions, automatic parking reminders, payment calculation, zone-specific pricing, and parking regulation notifications.

C. Municipality Dashboard

- Create parking zones.
- Update pricing.
- View parking activity.
- Analyze demand.
- Monitor parking revenue.
- Identify high-demand locations.

5. Traffic, Accident & Urban Mobility Map

To expand ParkFlow beyond a payment application, we designed an additional mobility component.

A. Live Traffic Map

- Traffic congestion.
- Accidents.
- Road closures.
- Road availability.
- Parking zones.
- Available routes.

B. Route Recommendation

Before reaching the parking location, the user can compare possible routes. The system can recommend a route based on estimated travel time, traffic level, reported incidents, road closures, and distance to the destination.

C. Parking + Mobility Integration

Instead of solving only the final parking step, ParkFlow aims to support the complete driver journey: Navigate -> Avoid Traffic -> Reach Parking Zone -> Park -> Automatically Pay -> Leave. This transforms ParkFlow from a simple parking-payment system into a broader Smart Urban Mobility Platform.

6. GitHub Repository & Development Management

A dedicated GitHub repository has been created for ParkFlow to organize the technical implementation and document the project's development.

A. Source Code Management

- Maintain the project source code.
- Track development progress.
- Store different application components.
- Manage changes using Git.
- Maintain a clear history of project development.

B. Project Documentation

- Project overview.
- Problem statement.
- Architecture.
- Installation instructions.
- Application setup.
- API structure.
- Database structure.
- Testing instructions.

C. Team Collaboration

- Work on separate features.
- Review code changes.
- Track issues.
- Organize development tasks.
- Maintain a reliable project history.

D. Technical Evidence

- Code implementation.
- Development progress.
- Architecture decisions.
- Testing.
- Iterative improvements.

7. Technical Feasibility & Prototype Planning

Several architecture alternatives were evaluated before choosing the current ParkFlow direction.

A. Technologies Evaluated

- GPS.
- Geofencing.
- Bluetooth.

- Mobile sensors.
- License plate recognition.
- Cameras.
- Parking sensors.
- Offline storage.
- Cloud/backend processing.

B. Cost-Oriented Architecture

A major project objective is to avoid requiring expensive infrastructure for every parking location. Therefore, ParkFlow follows a software-first architecture, with hardware used only when it adds sufficient value.

C. Progressive Deployment

- Mobile parking and digital payment.
- Parking-zone geofencing.
- Municipality management platform.
- Traffic and incident map.
- Additional parking verification mechanisms.
- Data analytics and intelligent parking recommendations.

8. Real-World Validation & Risk Analysis

The project was reviewed from the perspective of real-world driver behavior rather than only ideal technical scenarios.

A. Scenarios Studied

- User has no internet connection.
- User disables Bluetooth.
- Vehicle does not support Bluetooth.
- GPS accuracy is temporarily weak.
- User stops near parking without actually parking.
- User attempts to avoid paying.
- Parking location has no camera.
- Municipality cannot install expensive infrastructure.

B. Design Response

- Offline session support.
- Multi-signal parking verification.
- Reduced dependency on cameras.
- Reduced dependency on Bluetooth.
- Server-side validation.
- Progressive municipality infrastructure.

C. Practicality

The project continues to prioritize solutions that can realistically be deployed instead of features that work only under controlled prototype conditions.

9. Business & Sustainability Study

A. Target Customers

- Municipalities.
- Smart cities.
- Universities.
- Hospitals.
- Shopping centers.
- Private parking operators.
- Drivers.

B. Potential Revenue Models

- Small transaction fee per parking session.
- Municipality licensing.
- Monthly platform subscription.
- Private parking management contracts.
- Premium driver subscription.
- Analytics services.

C. Municipality Value

- Reduced dependency on parking meters.
- Lower infrastructure cost.
- Better parking revenue tracking.
- Digital parking records.
- Parking-demand analytics.
- Improved user experience.

D. Driver Value

- Easier parking payment.
- Payment based on actual parking time.
- Less dependency on physical parking meters.
- Traffic information.
- Route assistance.
- Digital parking history.

10. Current Project Progress

The project has progressed from an initial parking-payment concept into a broader technical architecture. Work completed so far includes:

- Parking problem research.
- Existing smart parking system research.
- Evaluation of parking-meter alternatives.
- Automatic parking-payment concept.
- GPS and geofencing study.
- Offline parking-session study.
- Bluetooth verification analysis.
- Camera-based verification analysis.
- Anti-fraud scenario analysis.
- Multi-signal parking verification architecture.

- Smart parking-zone architecture.
- Traffic and accident map concept.
- Route recommendation concept.
- Business-model study.
- Sustainability analysis.
- Real-world edge-case analysis.
- Technical feasibility analysis.
- GitHub repository creation.
- Project documentation planning.
- Prototype and implementation planning.

11. Next Implementation Phase

The next phase will focus on converting the completed research and architecture into a demonstrable working prototype.

A. Mobile Prototype

The application will demonstrate the main user flow: Open App -> View Map -> Reach Parking Zone -> Start Parking -> Track Duration -> Calculate Cost -> End Session.

B. Backend Integration

- User data.
- Parking zones.
- Parking sessions.
- Pricing.
- Transactions.

C. Map Prototype

- Parking locations.
- Traffic conditions.
- Accident or incident locations.
- Route options.

D. Testing

- Normal parking.
- Entering and leaving a parking zone.
- Short GPS interruptions.
- Internet disconnection.
- Invalid parking sessions.
- Repeated or suspicious parking behavior.

12. Project Vision

ParkFlow is being developed as more than a replacement for a traditional parking meter.

The long-term vision is to create a unified Smart Parking and Urban Mobility Platform that connects drivers, municipalities, parking infrastructure, payment systems, and mobility data.

The final experience aims to become: Drive -> Navigate -> Avoid Congestion -> Find Parking -> Park -> Automatically Pay -> Continue Your Journey, with minimum user interaction and minimum infrastructure cost.